

Alonso Marcos Muñoz

Data Engineer

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SUMMARY

Data Engineer with almost 2 years of experience in ETL pipelines, PostgreSQL and Oracle modelling and production deployment. Part of the ImpulsaDATA platform team at Tragsatec (15 people in a project of around 80), serving 22 Spanish ministries and public bodies. MSc in Big Data and Cloud Computing with hands-on Databricks, Spark, Kafka, Airflow and AWS.

EXPERIENCE

Tragsatec

Albacete, Spain

Data Engineer

Oct. 2025 - Present

- Build and maintain Python and SQL ETL pipelines that collect, validate and publish ImpulsaDATA catalogues daily or weekly, in a programme that has released 5,771 open datasets.
- Design PostgreSQL and Oracle transformations, aggregations and views so that data arrives structured, consistent and ready to be queried and published.
- Evolve the relational metadata model with the team (entities, relationships, states and vocabularies), protecting data integrity through every schema change.
- Automate the checks that validate every dataset against the European open data standard before release, keeping errors out of official catalogues.
- Set up and run development, test, pre-production and production environments with Docker, Podman, Git and SSH, keeping deployments documented, repeatable and easy for the team to review.
- Within the AgoraData project, lead development of the web interface designed for the data office (CDO, data owners and stewards) to operate ImpulsaDATA's backend and Python processes.

Tragsatec

Albacete, Spain

Data Engineer (Internship)

Jun. 2025 - Sep. 2025

- Developed the first version of the project's metadata conversion tool, a Python ETL that automatically publishes in CKAN public data that was not previously available as open data.

La Fábrica del Tiempo

Albacete, Spain

Data & Automation Engineer (Internship)

Feb. 2024 - Aug. 2024

- Improved the efficiency of lead management, notifications and approvals by more than 25%, automating them with Power Automate across SharePoint, the CRM and Microsoft 365.
- Delivered an internal Power Apps application that centralises the company's process management and information for a team of 10+ people.

PROJECTS

- **Databricks Telco Churn Lakehouse:** Created an end-to-end data platform that predicts telecom customer churn from 16 million synthetic records of 2 million customers, using PySpark, Delta Lake, Unity Catalog and MLflow: <https://github.com/alonsomarcosm99/databricks-telco-churn-lakehouse>
- **Batch and streaming platform:** Implemented a batch and real-time data platform with Spark, Kafka and Airflow that organises data into layers of increasing quality (Medallion architecture) on Delta Lake: <https://github.com/alonsomarcosm99/spark-kafka-airflow-data-platform>
- **AWS Smart Parking:** Deployed an AWS IoT platform that delivers data from 40 simulated parking sensors to its API in under 2 seconds at an estimated cost below USD 5 per month, using Lambda, DynamoDB and MQTT: <https://github.com/alonsomarcosm99/smart-parking-albacete>

EDUCATION AND CERTIFICATIONS

UCLM - MSc Big Data and Cloud Computing

2025 - 2026

- Spark, Databricks, streaming, AWS, machine learning, IoT and data governance. Master's thesis graded 9.2/10.

UCLM - BSc Computer Engineering, Information Technology

2019 - 2025

IES Juan Bosco - Higher Diploma in Networked Systems Administration

2017 - 2019

CAPM, project management certification - Project Management Institute

2025

B2 First - Cambridge English

2025

SKILLS

Data Engineering: SQL, Python, ETL and ELT pipelines, data modelling, PostgreSQL, Oracle, REST APIs and data quality.

Applied Big Data and cloud: Databricks, Spark, PySpark, Airflow, AWS, Kafka, Delta Lake, MLflow and Terraform.

Platforms: CKAN, OpenMetadata, DCAT-AP-ES, RDF, JSON-LD, SHACL, Power Apps and Power Automate.

Development and operations: Git, GitLab CI/CD, Docker, Podman, Linux, SSH, Java, Spring Boot and Vaadin.

Languages: native Spanish and professional English (B2 First).